ML-SUPERB Benchmark: Implementation and Extensions

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https://github.com/paulrgs21/ml-superb

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1 Introduction

ML-SUPERB [11] is a benchmark designed to evaluate Self-Supervised Learning (SSL) models on speech processing tasks. It extends the SUPERB benchmark to multilingual evaluation, as SUPERB was primarily focused on the English language. ML-SUPERB evaluates models across 143 different languages, ranging from high-resource to endangered ones.

The framework uses frozen SSL representations as input features for lightweight downstream models trained to perform various tasks, such as Automatic Speech Recognition (ASR) and Language Identification (LID). ASR is evaluated in both monolingual and multilingual settings. The multilingual ASR and LID tasks provide meaningful ways to compare SSL models and assess their ability to handle multilingual tasks. In particular, the framework makes important choices regarding the way SSL features are used, the datasets selected for fine-tuning, and the tasks chosen for evaluation.

Building on this framework, our project focuses on several key aspects of multilingual speech representation learning within ML-SUPERB. We first reimplement the monolingual ASR baseline using frozen SSL representations and a lightweight downstream model, ensuring a consistent evaluation setup. We then compare monolingual and multilingual pretrained models on a common ASR task to assess the impact of pretraining data diversity. The project also explores Parameter-Efficient Fine-Tuning (PEFT) methods, such as LoRA and adapters, as an alternative to frozen representations. We further investigate downstream architecture scaling, varying the capacity of the downstream model from a simple linear projection to multi-layer Transformer decoders, to assess how architectural choices interact with SSL encoder quality. In addition, we study finetuning strategies by contrasting monolingual and multilingual finetuning. Finally, we extend the evaluation to the Language Identification task, and analyze how encoder performance varies depending on their architecture and pre-training data.

2 Methodology

2.1 Evaluated Tasks

Monolingual ASR. As a baseline experiment, we focus on a low-resource language, Swahili. The dataset is extracted from the Common Voice [1] corpus, and we restrict ourselves to the 10-minute training subset due to computational constraints. The model is trained to predict character-level transcriptions using a CTC loss, and performance is measured using Character Error Rate (CER).

Language Identification. We also evaluate frozen SSL representations on a three-way language identification task. Following a simpler setup than the full ML-SUPERB LID benchmark, we train a linear decoder on top of frozen encoder representations to classify utterances among three low-resource African languages: Amharic, Swahili, and Wolof. The training data consists of the 10-minute subset of the ALFFA [7] corpus. The use of low-resource languages ensures the task is sufficiently challenging to differentiate encoder quality, and the choice of ALFFA guarantees that the evaluation data is not part of any model's pretraining set.

2.2 Foundation Models

We consider several upstream self-supervised speech models to cover a variety of architectures and pretraining conditions. As monolingual baselines, we use wav2vec2-base [3] and HuBERT-base [9]. On the multilingual side, we evaluate XLSR-53 [6], XLS-R-300M [2], and mHuBERT-147 [5], three models of different sizes trained on increasingly diverse language sets. In particular, mHuBERT-147 is a multilingual HuBERT-based model trained over three iterations on 147 languages, containing 95M parameters, making it relatively lightweight compared to other competitive SSL models. Although not included in the original ML-SUPERB benchmark, it serves as our primary model for the ASR experiments. The five models are summarized in Table 1.

Due to computational constraints, not all models could be evaluated on every experiment. However,

three core models, wav2vec2-base, HuBERT-base, and mHuBERT-147 are consistently tested across all tasks, providing a reliable basis for comparison between monolingual and multilingual pretraining.

Model	Type	Langs	Data
wav2vec2-base	mono	1	∼1k h
HuBERT-base	mono	1	∼60k h
XLSR-53	multi	53	∼56k h
XLS-R-300M	multi	128	∼436k h
mHuBERT-147	multi	147	∼90k h

Table 1: Summary of SSL models used in this study.

2.3 Downstream Model

As downstream architecture, we adopt the model proposed in SUPERB [12], with adaptations for the ASR task. First, a weighted sum of frozen SSL representations is computed using learnable layer-wise weights. SpecAugment is then applied, followed by a convolutional downsampling layer that reduces the temporal resolution by a factor of two.

The resulting representations are fed into a Transformer encoder composed of two layers, with an attention dimension of 256, a feedforward dimension of 1024, and 8 attention heads. A dropout rate of 0.1 is used throughout the model. Training is performed using the Connectionist Temporal Classification (CTC) loss.

We use the Adam optimizer with a learning rate of 10^{-4} and a weight decay of 10^{-6}. The batch size is set to 8 with gradient accumulation over 4 steps.

Due to computational constraints, training in the monolingual setting is limited to 1,566 iterations, corresponding to approximately 522 epochs. Performance is evaluated using the Character Error Rate (CER), following the ML-SUPERB protocol.

This configuration is kept identical across all experiments unless otherwise specified. In particular, we explore the effect of varying the downstream architecture capacity in Section 3.3.

3 Experiments

3.1 Monolingual ASR Baseline

We first report results for the baseline monolingual ASR experiment on Swahili using the 10-minute training subset. We evaluate our three core SSL models under the same conditions: frozen representations with a 2-layer Transformer downstream model. The results are summarized in Table 2.

mHuBERT-147 significantly outperforms the two monolingual models, achieving a CER of 22.43% on the test set with a best validation loss of 1.017.

SSL Model	CER (%)
wav2vec2-base	78.24
HuBERT-base	36.06
mHuBERT-147	22.43

Table 2: CER on Swahili monolingual ASR (10-min, 2-layer Transformer).

This improvement can be attributed to its multilingual pretraining on 147 languages, including Swahili. HuBERT-base obtains a reasonable CER of 36.06%, while wav2vec2-base lags behind at 78.24%, suggesting that the HuBERT pretraining objective produces more transferable representations for this low-resource setting.

Notably, mHuBERT-147 outperforms the models reported in the original ML-SUPERB study, including larger architectures. This can be attributed to several factors: the multilingual pretraining covering 147 languages, the use of FAISS-based k-means clustering for efficient label assignment, and a multilingual batching strategy with upsampling that improves data utilization across languages.

As shown in Figure 1, training with mHuBERT-147 was conducted for a total of 163 minutes. Both training and validation losses decrease steadily, with the best validation loss reached around the 60th epoch. The corresponding checkpoint was selected for the final evaluation. Despite the extremely limited amount of training data (92 utterances), the model is able to learn meaningful representations, highlighting the effectiveness of pretrained SSL features in low-resource settings.

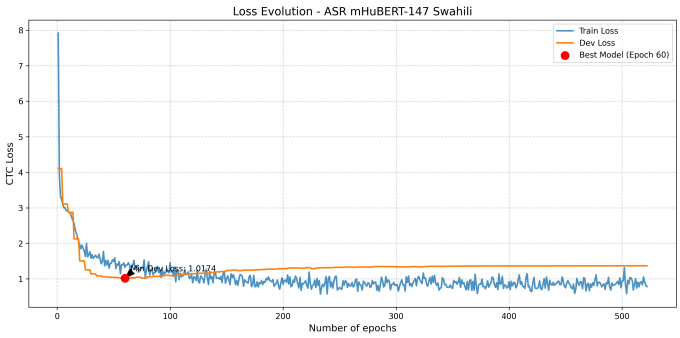


Figure 1: Training and validation loss curves for the Swahili 10-minute baseline experiment with mHuBERT-147.

We also analyzed the learned layer-wise weights of mHuBERT-147 used to combine SSL representations (Figure 2). The model assigns non-uniform importance across layers, indicating that different levels of abstraction contribute differently to the ASR task. As expected, the model places minimal weight on the initial CNN extractor and early Transformer layers, which primarily capture low-level acoustic fea-

tures. Instead, the importance peaks between layers 7 and 10. This concentration suggests that the most discriminative information for phonetic modeling and character recognition in mHuBERT-147 resides in these intermediate-to-late stages.

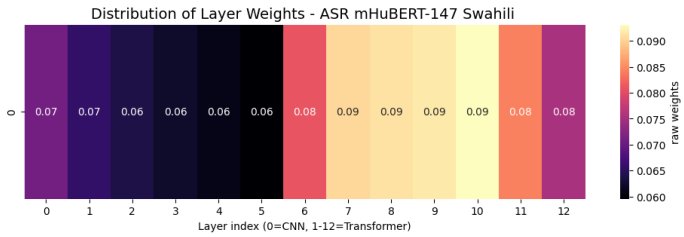


Figure 2: Layerwise weight analysis of mHuBERT-147 in the monolingual ASR baseline.

3.2 Parameter-Efficient Fine-tuning

The ML-SUPERB framework relies on fully frozen SSL representations. PEFT methods offer an alternative by introducing a small number of trainable parameters inside the encoder, allowing partial adaptation while preserving the pretrained weights.

We experiment with two methods: LoRA [10], which injects low-rank updates into the attention layers (r=8, α=16), and Houlsby adapters [8], which insert bottleneck modules (dimension 32) after each Transformer sub-layer. The parameter overhead is modest: from 9.09M (frozen) to 9.38M (LoRA) and 10.29M (Houlsby). Results are shown in Figure 3.

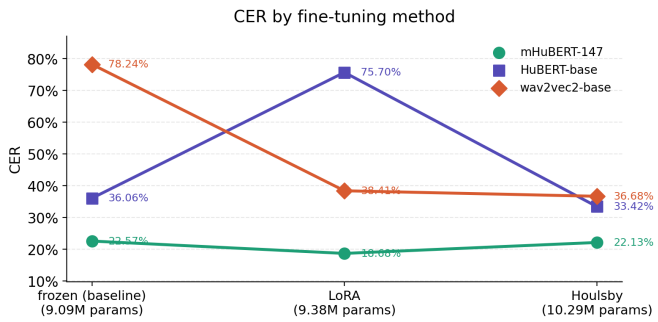


Figure 3: CER (%) on Swahili ASR (10-min) by fine-tuning method.

The effect of PEFT varies considerably across models. For mHuBERT-147, LoRA reduces the CER from 22.57% to 18.68%, confirming that well-aligned multilingual representations benefit from targeted low-rank adjustments. Houlsby adapters yield only a marginal gain.

For the monolingual models, it differs. On HuBERT-base, LoRA causes a severe degradation (75.70%), suggesting that low-rank updates disrupt representations not pretrained on the target language. Houlsby adapters, which inject capacity *between* layers rather than modifying attention weights, prove more stable with a modest improvement to 33.42%. On wav2vec2-base, both methods dramatically reduce the CER from 78.24% to around 37%, indicating that partial fine-tuning can recover useful features even from poorly matched representations.

These results suggest that the choice of PEFT strategy should be guided by the alignment between pretraining and downstream data: LoRA excels when representations are already suitable, while adapters offer a safer alternative otherwise.

3.3 Downstream Architecture Scaling

We investigate the impact of the downstream model's capacity on ASR performance by varying its architecture across three configurations: a simple linear projection (0.03M parameters), the baseline 2-layer Transformer (9.09M), and a larger 6-layer Transformer (24.85M). Results are shown in Figure 4.

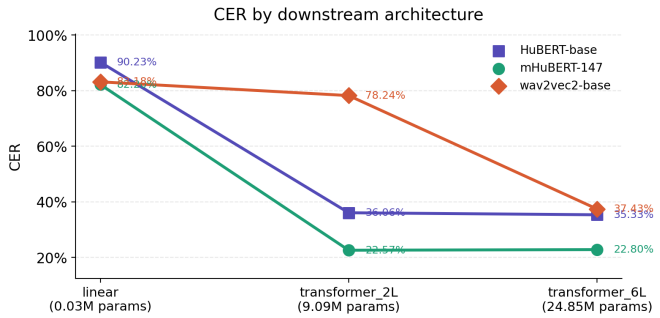


Figure 4: CER (%) on Swahili ASR (10-min) by downstream architecture.

With a linear head, all three models perform poorly (82–90% CER), confirming that a simple projection cannot decode character sequences from frozen SSL features. The jump to a 2-layer Transformer yields massive gains for mHuBERT-147 (22.57%) and HuBERT-base (36.06%), while wav2vec2-base remains high at 78.24%. Scaling to 6 layers brings diminishing returns for the first two models, but wav2vec2-base benefits substantially, dropping to 37.43% and reaching a CER comparable to HuBERT-base at 2 layers, but at the cost of nearly three times more trainable parameters.

This highlights a trade-off between encoder quality and downstream complexity: when SSL features are well-suited to the task, a lightweight downstream model suffices, but a more expressive architecture can partially compensate for weaker representations. Overall, the 2-layer Transformer chosen by ML-SUPERB appears well-calibrated: it captures most of the performance gains for strong encoders while remaining lightweight enough to keep the focus of the benchmark on the SSL representations rather than the downstream model.

3.4 Monolingual vs. Multilingual Fine-tuning

To assert whether or not the model could learn from many language to improve on one specific one, we extended the training data of the monolingual baseline. We tested to train for the ASR task with a sample of three languages including Swahili (swa, amh, wol) and on five languages (swa, amh, wol, ara, skr). Doing this, we extend the size of the training data. As these languages are not particularly close to the Swahili, the goal of the experiment is to access if more training data but in other languages can improve performances for a low-ressource language. We evaluated the three models on Swahili and we report the CER on the table bellow.

Number of Training Languages	**CER** (%)
1 (Swahili only)	22.43
3 (+ Amharic, Wolof)	26.88
4 (+ Arabic)	26.89
5 (+ Saraiki)	26.29

Table 3: Character Error Rate (CER) on Swahili test set as a function of the number of training languages. The monolingual baseline (1 language) is compared to multilingual fine-tuning configurations using mHuBERT-147 and 10-minute subsets.

The results show that performance decreases as more languages are added to the training set. However, we are not able to assess whether the number of languages itself is the determining factor. As the model is trained to perform an increasingly difficult task, it is not surprising that results worsen. This seems to suggest that for a fine-tuning task such as this one, data quality matters more than quantity. Further experiments could include typologically closer languages or a larger number of languages.

3.5 Language Identification

We reproduced the LID task from ML-SUPERB using a linear decoder on top of frozen SSL representations. We evaluated all five encoders described in Table 1 on this task.

We chose a simpler setup than the full ML-SUPERB LID benchmark in order to reduce the computational budget. Both mHuBERT-147 and XLS-R-300M were trained on data containing Amharic and Swahili, while Wolof is not included in any of their pretraining data. Therefore, this experiment may give us insight into the generalisation capacity of multilingual models to low-resource languages, and even to languages unseen during pretraining.

We find, in accordance with the results of [11], that monolingual models can perform surprisingly well compared to multilingual ones. Indeed, the best performances are achieved by wav2vec2-base, HuBERT-base and mHuBERT-147, all reaching above 90% accuracy. The two HuBERT models are difficult to distinguish from their final scores alone, which is why we include the training curves in Figure 6. We observe that training converged more easily for HuBERT-base than for its multilingual counterpart, possibly leading to better results on more challenging tasks. This appears consistent with the findings of [11], where HuBERT-base outperforms mHuBERT-base on the LID task.

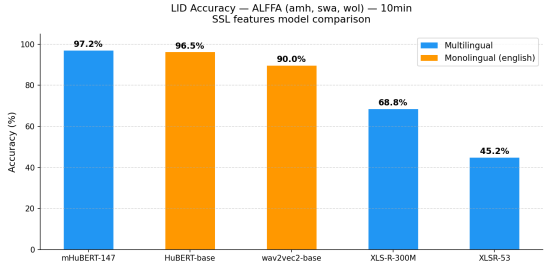
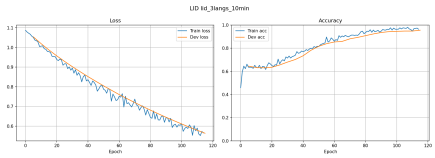
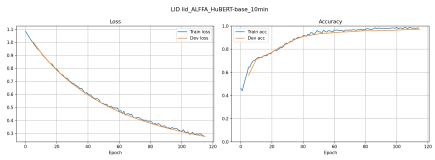


Figure 5: Accuracy comparison of 5 different encoders for the LID task.



(a) mHuBERT-147



(b) HuBERT-base

Figure 6: Training curves for LID on ALFFA.

4 Conclusion

This work highlights the importance of multilingual pretraining for low-resource ASR, with mHuBERT-147 consistently outperforming monolingual baselines. While PEFT methods can further improve performance, their effectiveness depends strongly on representation alignment. Surprisingly, monolingual models perform well on the LID task; in particular, HuBERT-base achieves performance comparable to mHuBERT-147, despite its pre-training data not containing any of the evaluated languages, while mHuBERT-147 was explicitly trained on some of them. Overall, results confirm that encoder quality remains the primary driver of performance in the ML-SUPERB framework.

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